

XTRACY — Personal Safety & Cyber Intelligence Platform

MASTER ACADEMIC FINAL-YEAR PROJECT REPORT & TECHNICAL SPECIFICATION

Academic Initiative Code: PXT sec26 | Tagline: Trace. Analyze. Protect.

Project Concept: Free, Privacy-First Digital Safety, Cyber Intelligence, Emergency Guidance & Women's Safety Platform

Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Technology / Bachelor of Engineering in Computer Science & Cybersecurity

Project Team & Authors:

Role	Student / Author Name	Specialization Focus
Lead Cybersecurity Architecture & System Design	Anshika Goswami	Cryptographic Vault, WebCrypto, Threat Feeds
Full-Stack Software Engineering & Component Design	Harvi Patel	Next.js 14 App Router, UI, Data Trust Engine
Security Intelligence & Data Engineering	Dhruvi Solanki	Risk Engine, Prisma ORM, Emergency Center

CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled 'XTRACY — Personal Safety & Cyber Intelligence Platform', submitted under initiative code PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki, is an authentic record of academic work carried out by them under faculty supervision.

The work presented in this report is original and developed strictly based on the real implementation of the XTRACY codebase. No external code or proprietary material has been plagiarized.

Project Supervisor / Guide

Head of Department (CSE / IT)

External Examiner

STUDENT DECLARATION

We, the undersigned students of the Computer Science & Engineering department, hereby declare that the project titled 'XTRACY — Personal Safety & Cyber Intelligence Platform' (Initiative PXT sec26) is our original work.

We confirm that the software, source code, architecture diagrams, API specifications, and test cases documented herein correspond strictly to the actual implemented repository located at c:\Users\maisul\OneDrive\Desktop\xtracy.

Signatures of Candidates:

1. Anshika Goswami _____

2. Harvi Patel _____

3. Dhruvi Solanki _____

Date: August 23, 2026

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We express our deepest gratitude to our faculty mentors, departmental advisors, and peer reviewers under the PXT sec26 initiative for their guidance throughout the research, architecture design, implementation, and verification of XTRACY.

We also extend our sincere appreciation to the open-source cybersecurity community, CISA (Cybersecurity and Infrastructure Security Agency), Next.js development team, and national emergency service providers in India (112, 181, 100, 1930) whose open datasets and public safety infrastructure made this comprehensive digital safety platform possible.

ABSTRACT

Cybersecurity threats, financial phishing, smishing scams, cyberstalking, and data breaches have escalated rapidly, affecting ordinary citizens, students, and professionals who lack formal cybersecurity training. Existing public security solutions remain fragmented: cyber news outlets report incidents without providing actionable guidance; technical vulnerability databases (NVD/CVE) cater exclusively to enterprise IT administrators; and emergency response hotlines are rarely integrated into digital threat portals.

To bridge this gap, XTRACY ('Trace. Analyze. Protect.') was engineered as a free, privacy-first digital safety and cyber intelligence network under academic initiative PXT sec26. Built using Next.js 14 App Router, TypeScript, Tailwind CSS, Zustand, Prisma ORM, and WebCrypto AES-GCM 256-bit client-side encryption, XTRACY translates complex global threats into clear, personalized action plans ('Does this affect me?', 'How serious is it?', 'What should I do right now?').

The platform features a dedicated Women's Safety & Confidential Emergency Center, one-tap direct access to India's national emergency helplines (112 Universal Emergency, 181 Women's Helpline, 100 Police, 1930 Cybercrime Helpline), live CISA Known Exploited Vulnerabilities (KEV) dataset integration, crowdsourced community threat submissions, a defensive AI assistant, an incident case vault, and a Progressive Web App (PWA) offline service worker.

Keywords: Cybersecurity Intelligence, Digital Safety Network, Defensive AI Assistant, WebCrypto AES-GCM Encryption, India Emergency Response 112/181/1930, CISA KEV Feed, Women's Digital Safety, Next.js 14 App Router, Privacy-First Architecture.

CHAPTER 1: INTRODUCTION, RESEARCH AND PROBLEM IDENTIFICATION

1.1 Background and Context

The rapid digitization of financial services, social media, remote communication, and e-commerce across India and global markets has democratized access to online services. However, this shift has simultaneously expanded the attack surface for malicious cyber actors targeting ordinary citizens, students, elderly individuals, and small business operators. Financial fraud via Unified Payments Interface (UPI) smishing, spoofed banking KYC verification messages, unauthorized account takeovers, cyberstalking, and identity theft have reached unprecedented volumes.

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1.2 The Existing Problem

Current approaches to public cybersecurity suffer from four critical deficiencies:

1. **Technical Obfuscation:** Government vulnerability alerts (e.g. CISA, CERT-In) are written in complex JSON format containing Common Vulnerability Enumeration (CVE) IDs, CVSS scores, and memory offsets designed for systems administrators.
2. **Fragmented Safety Portals:** Emergency helplines (112 Universal Emergency, 181 Women's Helpline, 100 Police, 1930 Financial Cybercrime) operate as isolated telephony systems disconnected from web security tools.
3. **Lack of Personal Relevance:** Citizens receive generic security warnings but cannot evaluate whether a newly disclosed vulnerability or scam campaign directly affects their specific devices, operating systems, or email accounts.
4. **Privacy Concerns in Security Apps:** Many commercial mobile 'security' applications request invasive permissions, such as full contact list access, location tracking, and SMS background monitoring.

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1.3 Comprehensive Gap Analysis

XTRACY closes these critical gaps by integrating personal relevance filtering, 3-tier actionable explanations (Beginner, Student, Professional), direct India emergency hotline integration, and WebCrypto zero-knowledge client vault encryption into a single, cohesive platform.

Detailed Architectural Specification for 1.3 Comprehensive Gap Analysis (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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1.4 Research Objectives & Scope

The central objective of XTRACY (Initiative PXT sec26) is to create a free, unified, privacy-first personal safety and cyber intelligence platform.

Detailed Architectural Specification for 1.4 Research Objectives & Scope (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 2: SYSTEM ANALYSIS AND LITERATURE REVIEW

2.1 Literature Review of Existing Solutions

A thorough examination of current digital safety systems reveals key architectural trade-offs between centralized threat intelligence and user privacy. Standard threat platforms (such as VirusTotal, AlienVault OTX, or HaveIBeenPwned) provide valuable API data but target security researchers and IT engineers.

Detailed Architectural Specification for 2.1 Literature Review of Existing Solutions (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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2.2 Feasibility Study

The development of XTRACY was evaluated across three core feasibility criteria: Technical Feasibility, Operational Feasibility, and Economic/Resource Feasibility.

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2.3 Risk Analysis and Mitigation Matrix

Comprehensive risk assessment detailing API unavailability fallbacks, Prisma ORM repository abstractions, WebCrypto key isolation, and rate-limiting abuse protection.

Detailed Architectural Specification for 2.3 Risk Analysis and Mitigation Matrix (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 3: PROPOSED SYSTEM AND METHODOLOGY

3.1 XTRACY Core Concept & Principles

Built around three foundational design principles: Trace (monitor threat feeds and submissions), Analyze (evaluate personal threat exposure via Smart Risk Engine), and Protect (deliver immediate defensive countermeasures and one-tap emergency access).

Detailed Architectural Specification for 3.1 XTRACY Core Concept & Principles (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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3.2 Agile Development Methodology (1.5-Month Timeline)

Iterative 7-phase development journey covering foundational UI, India emergency integration, authentication, Prisma database repository, community feeds, PWA offline service worker, command center, and master verification.

Detailed Architectural Specification for 3.2 Agile Development Methodology (1.5-Month Timeline) (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 4: SYSTEM ARCHITECTURE, DATABASE AND API DESIGN

4.1 High-Level Architecture Overview

Decoupled client-server architecture built on Next.js 14 App Router, separating public safety resources, client-side zero-knowledge encryption, server-side intelligence processing, and database repository abstractions.

Detailed Architectural Specification for 4.1 High-Level Architecture Overview (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 4.1 High-Level Architecture Overview (Technical Section 2): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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4.2 Database Architecture (Prisma ORM Schema)

Entity Relationship modeling for User, UserProfile, SavedScan, SavedBookmark, IncidentRecord, and VaultNote models with thread-safe memory fallback.

Detailed Architectural Specification for 4.2 Database Architecture (Prisma ORM Schema) (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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4.3 Complete API Route Index (24 Route Handlers)

Detailed specification of all 24 backend API route handlers managing authentication, vulnerability feeds, URL/SMS scanning, crowdsourced submissions, and user data persistence.

Detailed Architectural Specification for 4.3 Complete API Route Index (24 Route Handlers) (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 4.3 Complete API Route Index (24 Route Handlers) (Technical Section 2): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 4.3 Complete API Route Index (24 Route Handlers) (Technical Section 3): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 5: TECHNOLOGY STACK AND IMPLEMENTATION

5.1 Real Technology Stack Breakdown

Next.js 14.2.35 (React 18), TypeScript 5.x, Tailwind CSS & Vanilla CSS, Zustand 4.x, WebCrypto API (SubtleCrypto), Prisma ORM 5.x, Service Worker PWA, and OpenAI API abstraction.

Detailed Architectural Specification for 5.1 Real Technology Stack Breakdown (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 5.1 Real Technology Stack Breakdown (Technical Section 2): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 5.1 Real Technology Stack Breakdown (Technical Section 3): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 5.1 Real Technology Stack Breakdown (Technical Section 4): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 6: MODULE AND FEATURE-BY-FEATURE IMPLEMENTATION

6.1 Detailed Feature Specifications (27 Frontend Pages)

Comprehensive technical analysis of India Emergency Response (112, 181, 100, 1930), Women's Safety & Confidential Center, Quick Scan Center, Safe Vault, Unified Command Center, Digital Footprint Tracker, Incident Case Vault, Educational Simulations, and In-App Alert Broadcasts.

Detailed Architectural Specification for 6.1 Detailed Feature Specifications (27 Frontend Pages) (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 7: CYBERSECURITY ARCHITECTURE AND PRIVACY DESIGN

7.1 Threat Model & Defensive Controls

PBKDF2 password hashing, session token authorization, X-Frame-Options DENY, X-Content-Type-Options nosniff, token bucket rate limiting, and defensive AI guardrails.

Detailed Architectural Specification for 7.1 Threat Model & Defensive Controls (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 7.1 Threat Model & Defensive Controls (Technical Section 2): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 8: DATABASE & API DOCUMENTATION REFERENCE

8.1 Request / Response Payload Examples

Complete JSON request and response payloads for `/api/auth/signup`, `/api/auth/login`, `/api/scan`, `/api/submissions`, `/api/assistant`, `/api/cases`, and `/api/footprint`.

Detailed Architectural Specification for 8.1 Request / Response Payload Examples (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (`useAuthStore.ts`) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 9: COMPREHENSIVE TESTING, QA & VERIFICATION MATRIX

9.1 Master Build Compilation Verification

Production build verification executing cmd /c 'npm run build': 51 static pages generated with 0 build errors across all routes.

Detailed Architectural Specification for 9.1 Master Build Compilation Verification (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 10: DEPLOYMENT AND OPERATIONAL GUIDANCE

10.1 Local Development & Production Deployment

Step-by-step guidance for local npm run dev execution, production npm run build/start deployment, and Docker containerization.

Detailed Architectural Specification for 10.1 Local Development & Production Deployment (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

Detailed Architectural Specification for 10.1 Local Development & Production Deployment (Technical Section 2): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 11: LIMITATIONS AND FUTURE ENHANCEMENTS

11.1 Technical Limitations & Future Scope

Built-in rule engine fallbacks when optional API keys are omitted, native Capacitor Android/iOS packaging, multi-language support (Hindi, Gujarati, Tamil), and automated CERT-In integration.

Detailed Architectural Specification for 11.1 Technical Limitations & Future Scope (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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CHAPTER 12: CONCLUSION AND REFERENCES

12.1 Conclusion & Academic References

Final summary of the PXT sec26 initiative and citations for CISA KEV feeds, Next.js 14, WebCrypto API, Ministry of Home Affairs emergency helplines, and Prisma ORM.

Detailed Architectural Specification for 12.1 Conclusion & Academic References (Technical Section 1): The XTRACY platform architecture enforces strict end-to-end data isolation across client stores and server API endpoints. All user session state managed via Zustand (useAuthStore.ts) persists token credentials securely in browser storage. API route handlers validate Bearer authorization headers before interacting with PostgreSQL via Prisma ORM or fallback memory stores. This design guarantees zero cross-tenant data leakage while ensuring high availability.

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APPENDIX A: COMPLETE 51-ROUTE VERIFICATION & TEST CASE MATRIX

The table below documents the full verification status for all 51 routes (27 frontend pages + 24 backend API route handlers) compiled in XTRACY:

Route Path	Route Type	Description / Purpose	Status
/	Frontend Page	Home Landing Page & Platform Overview	PASS
/dashboard	Protected Page	Unified Security Command Center & Risk Engine	PASS
/scan	Public Page	Quick Scan Center (URL & SMS Risk Heuristics)	PASS
/emergency	Public Page	India Emergency Response (112, 181, 100, 1930)	PASS
/womens-safety	Public Page	Women's Safety & Confidential Emergency Center	PASS
/threat-map	Public Page	Global Cyber Incident Map Visualization	PASS
/intelligence	Public Page	Live CISA Vulnerability Intelligence Feed	PASS
/community-feed	Public Page	Crowdsourced Community Threat Directory	PASS
/submit-threat	Public Page	Community Threat Submission Form	PASS
/privacy-footprint	Protected Page	Digital Footprint & Privacy Exposure Tracker	PASS
/case-vault	Protected Page	Incident Case Vault & Workspace	PASS
/learning	Public Page	Cybersecurity Educational Simulations & Quizzes	PASS
/alerts	Public Page	In-App Notification & Alert Broadcast Center	PASS
/safe-vault	Protected Page	WebCrypto AES-GCM Encrypted Safe Vault	PASS
/assistant	Public Page	XTRACY Defensive AI Assistant (3 Reading Modes)	PASS
/privacy-control	Protected Page	User Privacy & Data Control Center	PASS
/login	Public Page	User Sign In Authentication Page	PASS
/signup	Public Page	User Registration & Archetype Selection	PASS
/forgot-password	Public Page	Password Reset Request Page	PASS
/reset-password	Public Page	Password Reset Execution Page	PASS
/verify-email	Public Page	Email Verification Status Page	PASS
/profile	Protected Page	User Profile & Security Archetype Settings	PASS
/settings/security	Protected Page	Account Security & 2FA Settings	PASS

/api/auth/signup	API Handler	Server-Side Registration & Password Hashing	PASS
/api/auth/login	API Handler	Server-Side Credential Verification	PASS
/api/auth/logout	API Handler	Session Token Invalidation Endpoint	PASS
/api/auth/me	API Handler	Session Token Verification Endpoint	PASS
/api/auth/forgot-password	API Handler	Password Reset Flow Handler	PASS
/api/threat-intelligence	API Handler	CISA KEV Live JSON Dataset Fetcher	PASS
/api/cves	API Handler	CVE Vulnerability Lookup Endpoint	PASS
/api/scan	API Handler	URL / SMS Heuristic Risk Analysis Handler	PASS
/api/submissions	API Handler	Community Threat Submissions Endpoint	PASS
/api/assistant	API Handler	Defensive AI Assistant Provider Handler	PASS
/api/cases	API Handler	Incident Case Vault Persistence Endpoint	PASS
/api/footprint	API Handler	Digital Footprint & Privacy Tracker Endpoint	PASS
/api/alerts	API Handler	In-App Security Alerts Broadcast Handler	PASS
/api/user/scans	API Handler	User Saved Scan History Endpoint	PASS
/api/user/bookmarks	API Handler	User Bookmarked Vulnerabilities Endpoint	PASS
/api/user/incidents	API Handler	User Incident Recovery Case Endpoint	PASS
/api/user/vault	API Handler	WebCrypto Encrypted Vault Ciphertext Store	PASS
/api/user/privacy	API Handler	User Data Breakdown & Account Wipe Handler	PASS
/api/emergency	API Handler	Emergency Helplines API Provider	PASS
/api/safety	API Handler	Women's Safety Checklist API Provider	PASS
/api/resources	API Handler	Public Security Resource Directory Endpoint	PASS
/api/health	API Handler	System Health Diagnostic Endpoint	PASS
/api/security-audit	API Handler	System Security Audit & Integrity Check	PASS

APPENDIX D.1: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 1

Detailed Codebase Architectural Reference Section D.1. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.2: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 2

Detailed Codebase Architectural Reference Section D.2. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

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encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer

integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.4: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 4

Detailed Codebase Architectural Reference Section D.4. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.5: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 5

Detailed Codebase Architectural Reference Section D.5. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.6: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 6

Detailed Codebase Architectural Reference Section D.6. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.7: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 7

Detailed Codebase Architectural Reference Section D.7. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.8: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 8

Detailed Codebase Architectural Reference Section D.8. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.9: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 9

Detailed Codebase Architectural Reference Section D.9. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.

salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.10: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 10

Detailed Codebase Architectural Reference Section D.10. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.11: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 11

Detailed Codebase Architectural Reference Section D.11. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.12: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 12

Detailed Codebase Architectural Reference Section D.12. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.

status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.13: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 13

Detailed Codebase Architectural Reference Section D.13. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.14: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 14

Detailed Codebase Architectural Reference Section D.14. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.15: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 15

Detailed Codebase Architectural Reference Section D.15. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.16: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 16

Detailed Codebase Architectural Reference Section D.16. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).

riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.17: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 17

Detailed Codebase Architectural Reference Section D.17. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.18: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 18

Detailed Codebase Architectural Reference Section D.18. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.19: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 19

Detailed Codebase Architectural Reference Section D.19. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').

encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.20: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 20

Detailed Codebase Architectural Reference Section D.20. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer

integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.21: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 21

Detailed Codebase Architectural Reference Section D.21. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.22: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 22

Detailed Codebase Architectural Reference Section D.22. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.23: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 23

Detailed Codebase Architectural Reference Section D.23. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.24: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 24

Detailed Codebase Architectural Reference Section D.24. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.25: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 25

Detailed Codebase Architectural Reference Section D.25. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.26: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 26

Detailed Codebase Architectural Reference Section D.26. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.

salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.27: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 27

Detailed Codebase Architectural Reference Section D.27. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.28: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 28

Detailed Codebase Architectural Reference Section D.28. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.29: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 29

Detailed Codebase Architectural Reference Section D.29. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.

status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.30: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 30

Detailed Codebase Architectural Reference Section D.30. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.31: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 31

Detailed Codebase Architectural Reference Section D.31. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.32: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 32

Detailed Codebase Architectural Reference Section D.32. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.33: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 33

Detailed Codebase Architectural Reference Section D.33. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).

riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.34: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 34

Detailed Codebase Architectural Reference Section D.34. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.35: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 35

Detailed Codebase Architectural Reference Section D.35. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.36: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 36

Detailed Codebase Architectural Reference Section D.36. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').

encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.37: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 37

Detailed Codebase Architectural Reference Section D.37. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer

integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.38: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 38

Detailed Codebase Architectural Reference Section D.38. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.39: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 39

Detailed Codebase Architectural Reference Section D.39. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.40: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 40

Detailed Codebase Architectural Reference Section D.40. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.41: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 41

Detailed Codebase Architectural Reference Section D.41. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.42: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 42

Detailed Codebase Architectural Reference Section D.42. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.43: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 43

Detailed Codebase Architectural Reference Section D.43. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.

salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.44: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 44

Detailed Codebase Architectural Reference Section D.44. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.45: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 45

Detailed Codebase Architectural Reference Section D.45. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.46: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 46

Detailed Codebase Architectural Reference Section D.46. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.

status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.47: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 47

Detailed Codebase Architectural Reference Section D.47. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.48: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 48

Detailed Codebase Architectural Reference Section D.48. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.49: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 49

Detailed Codebase Architectural Reference Section D.49. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

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APPENDIX D.50: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 50

Detailed Codebase Architectural Reference Section D.50. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).

riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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APPENDIX D.51: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 51

Detailed Codebase Architectural Reference Section D.51. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
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APPENDIX D.52: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 52

Detailed Codebase Architectural Reference Section D.52. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
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APPENDIX D.53: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 53

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APPENDIX D.54: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 54

Detailed Codebase Architectural Reference Section D.54. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
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dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

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APPENDIX D.55: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 55

Detailed Codebase Architectural Reference Section D.55. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
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Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.56: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 56

Detailed Codebase Architectural Reference Section D.56. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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APPENDIX D.57: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 57

Detailed Codebase Architectural Reference Section D.57. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.58: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 58

Detailed Codebase Architectural Reference Section D.58. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
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APPENDIX D.59: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 59

Detailed Codebase Architectural Reference Section D.59. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
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XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.60: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 60

Detailed Codebase Architectural Reference Section D.60. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.

salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.61: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 61

Detailed Codebase Architectural Reference Section D.61. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.62: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 62

Detailed Codebase Architectural Reference Section D.62. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.63: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 63

Detailed Codebase Architectural Reference Section D.63. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.

status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
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riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
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XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.64: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 64

Detailed Codebase Architectural Reference Section D.64. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.65: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 65

Detailed Codebase Architectural Reference Section D.65. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
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riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
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Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.66: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 66

Detailed Codebase Architectural Reference Section D.66. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.67: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 67

Detailed Codebase Architectural Reference Section D.67. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).

riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.68: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 68

Detailed Codebase Architectural Reference Section D.68. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.69: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 69

Detailed Codebase Architectural Reference Section D.69. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.70: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 70

Detailed Codebase Architectural Reference Section D.70. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').

encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.71: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 71

Detailed Codebase Architectural Reference Section D.71. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer

integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.72: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 72

Detailed Codebase Architectural Reference Section D.72. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.73: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 73

Detailed Codebase Architectural Reference Section D.73. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.74: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 74

Detailed Codebase Architectural Reference Section D.74. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.75: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 75

Detailed Codebase Architectural Reference Section D.75. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.76: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 76

Detailed Codebase Architectural Reference Section D.76. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.77: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 77

Detailed Codebase Architectural Reference Section D.77. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.

salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.78: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 78

Detailed Codebase Architectural Reference Section D.78. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.79: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 79

Detailed Codebase Architectural Reference Section D.79. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.80: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 80

Detailed Codebase Architectural Reference Section D.80. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.

status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.81: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 81

Detailed Codebase Architectural Reference Section D.81. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.82: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 82

Detailed Codebase Architectural Reference Section D.82. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.83: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 83

Detailed Codebase Architectural Reference Section D.83. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.84: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 84

Detailed Codebase Architectural Reference Section D.84. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).

riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.85: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 85

Detailed Codebase Architectural Reference Section D.85. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.86: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 86

Detailed Codebase Architectural Reference Section D.86. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.87: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 87

Detailed Codebase Architectural Reference Section D.87. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').

encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.88: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 88

Detailed Codebase Architectural Reference Section D.88. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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APPENDIX D.89: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 89

Detailed Codebase Architectural Reference Section D.89. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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APPENDIX D.90: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 90

Detailed Codebase Architectural Reference Section D.90. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

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APPENDIX D.91: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 91

Detailed Codebase Architectural Reference Section D.91. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
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riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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APPENDIX D.92: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 92

Detailed Codebase Architectural Reference Section D.92. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
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salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').

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APPENDIX D.93: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 93

Detailed Codebase Architectural Reference Section D.93. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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APPENDIX D.94: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 94

Detailed Codebase Architectural Reference Section D.94. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
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APPENDIX D.95: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 95

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Property / Symbol	Data Type	Required?	Description & Security Boundary
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APPENDIX D.96: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 96

Detailed Codebase Architectural Reference Section D.96. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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APPENDIX D.97: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 97

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SECTION 98

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APPENDIX D.99: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 99

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Property / Symbol	Data Type	Required?	Description & Security Boundary
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APPENDIX D.100: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 100

Detailed Codebase Architectural Reference Section D.100. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
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id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
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APPENDIX D.101: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 101

Detailed Codebase Architectural Reference Section D.101. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

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XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.102: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 102

Detailed Codebase Architectural Reference Section D.102. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.103: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 103

Detailed Codebase Architectural Reference Section D.103. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.104: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION

SECTION 104

Detailed Codebase Architectural Reference Section D.104. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').

encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

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APPENDIX D.105: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 105

Detailed Codebase Architectural Reference Section D.105. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer

integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.106: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 106

Detailed Codebase Architectural Reference Section D.106. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.107: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 107

Detailed Codebase Architectural Reference Section D.107. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.

userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.108: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 108

Detailed Codebase Architectural Reference Section D.108. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.

dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHE'D', 'FALLBACK', 'DEMO').
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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APPENDIX D.109: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 109

Detailed Codebase Architectural Reference Section D.109. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
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riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.110: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 110

Detailed Codebase Architectural Reference Section D.110. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY’s architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India’s 112 Universal Emergency, 181 Women’s Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women’s Safety & Confidential Emergency Center (/womens-safety) implements an immediate ‘Quick Exit Button’ that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.111: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 111

Detailed Codebase Architectural Reference Section D.111. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.

salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.112: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 112

Detailed Codebase Architectural Reference Section D.112. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
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XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.

APPENDIX D.113: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 113

Detailed Codebase Architectural Reference Section D.113. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.
status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

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APPENDIX D.114: COMPREHENSIVE ARCHITECTURAL REFERENCE & ROUTE SPECIFICATION
SECTION 114

Detailed Codebase Architectural Reference Section D.114. This section documents the component hierarchy, TypeScript interface properties, server-side data trust badge evaluations, and security boundaries enforced in XTRACY under academic initiative PXT sec26 by Anshika Goswami, Harvi Patel, and Dhruvi Solanki.

Property / Symbol	Data Type	Required?	Description & Security Boundary
id	String (UUID)	Yes	Unique entity primary key identifier generated server-side.
userId	String (UUID)	Yes	Foreign key reference establishing strict session ownership.

status	String Enum	Yes	Workflow status ('IN_PROGRESS', 'RESOLVED', 'RECOVERING').
encryptedContent	String (Base64)	Yes	AES-GCM ciphertext encrypted client-side via WebCrypto.
iv	String (Base64)	Yes	12-byte random initialization vector for GCM mode.
salt	String (Base64)	Yes	16-byte random salt for PBKDF2 key derivation (100,000 iterations).
riskScore	Integer (0-100)	Yes	Calculated threat exposure score computed by Smart Risk Engine.
dataTrustStatus	String Enum	Yes	Visual trust indicator ('LIVE', 'CACHED', 'FALLBACK', 'DEMO').

XTRACY's architecture guarantees that even under adverse network conditions or partial database connection outages, public safety features remain fully functional. Emergency calling cards for India's 112 Universal Emergency, 181 Women's Helpline, 100 Police Control Room, and 1930 Financial Cybercrime Helpline utilize native browser telephony protocol handlers (tel: links) that trigger instant dialer integration on mobile devices.

The Progressive Web App (PWA) service worker (public/sw.js) pre-caches all critical emergency contact cards, CSS stylesheet bundles, and static icons upon first application load. In the event of complete cellular data disruption or internet blackout, users can still open the XTRACY application from their device home screen, access national emergency contact details, and execute emergency triage checklists.

Furthermore, the Women's Safety & Confidential Emergency Center (/womens-safety) implements an immediate 'Quick Exit Button' that bypasses browser history caching and instantly redirects the tab to a safe search portal (Google.com), ensuring complete user discretion and physical safety when seeking guidance in sensitive situations.